

What did Human Ancestors Eat?

We know something about what our early ancestors ate based on primate diet, but detailed study of hominin fossils gives even greater insight into the diet of these extinct species.

Evidence from Evolution and Anatomy

Sometimes, however, ecological analogy is insufficient. For example, *Paranthropus boisei*, with its large grinding teeth, existed from c. 2.3 million to 1.2 million years ago in East Africa, in the same period and area as early species of *Homo* (e.g., *Homo habilis*, which existed c. 2.3–1.65 million years ago). Based on differences in dentition between these two species, it has been inferred that they ate distinct diets, since they resided in the same area. It is possible, however, that the species lived within different habitats in this landscape, which is why they evolved to focus on different foods. *Paranthropus boisei* shows an evolutionary adaptation to chewing, based on its dentition, but also evidence of what was chewed. Microwear on fossil molars suggests a diet high in abrasive foods or food inclusions (likely, sand), rather than one focused on large, hard objects, which was the original theory for the diet of this species, with its massive jaws (it was nicknamed “nutcracker man”!). Stable isotope analysis, which we will cover in detail in a later module, indicates that grasses and sedges were a much greater portion of its diet than were plants from forested areas.

The absence of evidence of tough foods in the diet of early *Homo*—together with its greater metabolic needs, inferred from the larger brain size of that lineage—has led researchers to suggest that meat made up a larger proportion of its diet than that of *Paranthropus boisei*. *Homo habilis*, for example, is thought to have been a scavenger, chasing off small predators from recent kills. This source of meat in large packages—together with increasingly complex stone tools used to skin and butcher animals, crush bones for marrow, and cut tough plant tissues—was an alternative adaptation to the resources available in East Africa at the time. At least some social consumption of meat may have occurred in this period, given the likelihood that some scavenged meat came in larger packages than were useable by any individual or family group.

Another clue is the chest structure of hominins and the digestive tract of our species (which we infer to be similar in other species of our genus, at least). We have smaller stomachs than other primates, indicating that we evolved to eat relatively nutrient-dense food in lower volumes than, for example, gorillas. Our intestinal length is relatively long to maximize nutrient absorption from these foods. Given that our genus has smaller teeth than those of our early hominin ancestors and cousins, these adaptations together indicate the role of technology in food processing, increasing nutrient density in foods and rendering them edible, with less work required from our jaws.

Evidence from Archaeology

How important animal versus plant foods were in the diet of early hominins, including early lineages of *Homo*, is still debated. Both the “Man the Hunter” and “Woman the Gatherer” models that associate key advents in human evolution to specific food items are useful **heuristic models**, but likely not entirely descriptive of the key dietary contributions during human evolution. Archaeology provides primary evidence, in the form of tools and the remains of food, that indicates the types of foods and strategies for acquiring that food practiced by hominins.

Heuristic models are theoretical models designed to facilitate thinking while being only partial or approximate. The emphasis of heuristic models is on thinking and prediction, rather than absolute accuracy. For example, Newton’s Laws of Physics are not heuristic, as they attempted to describe reality as accurately as possible. The scavenging hypothesis is an example of a heuristic model.

As described earlier, our digestive and masticatory (chewing) anatomy indicates adaptations to relatively nutrient-dense, low-mass foods. These adaptations likely could have occurred only in the context of tools use by our ancestors, including fire for cooking and tools for crushing, pounding, and grinding plant foods to render them easier to digest. While some forms of cooking are thus likely to have occurred in earlier periods, evidence increasingly suggests that the most clear-cut and sustained evidence for the use of fire, grinding tools, and other cooking technologies is associated with our species, *Homo sapiens*.